

the dream. From being an industry changing technology that would enable manufacturing to gain new heights, it has instead found its appeal in the hands of individuals, who embody the potential of 3D printing technology through early adoption and experimentation.

And even as 3D printing technologies continue to develop and mature, the benchmark for their potential is higher than can be imagined if we consider the underlying potential of the technology. We will take a look at the foundations of 3D printing technology and see how they connect to its applications in the present as well as the future.



Even surrealist imagery isn't impossible

Potential 1: Manufac-

turing with 3D printers has been proven to be elegant and simple when compared with traditional manufacturing methods. Typically we find that the more complex an objects design or shape it, the more expensive it is to produce, when using

traditional technologies. However with 3D printers, this problem is resolved, as the level of complexity has no bearing on the costs of manufacturing. Since the craftsmanship required is unified in the software end of the process, issues such as skill, time and cost are relatively the same across a wide variety of objects. The only variable becomes the cost of materials and time taken to fabricate, which is independent from the complexity of the design. This feature disrupts the traditional economics of how products are valued and changes the business of manufacturing radically.

Potential 2: The age of mass production has brought us to a place where it is easier and cheaper to manufacture identical objects in large units to optimize costs. But since 3D printing exists outside of this constraint, the potential for infinite variety is already at hand. A 3D printer is not restrained by the design of its products and can produce objects of a wide variety with equal ease. Traditional methods require fixed assets like moulds or assembly line configurations that are complicated and expensive to implement, making it worthwhile only if large units of identical products are manufactured, but with 3D printing we can create a variety of products using the same 3D printer asset. It effectively allows for an all-in-one factory.